

Nobleson Kunjappy

Postdoctoral Researcher

nobleson@kumamoto-u.ac.jp | ORCID: 0000-0003-2715-4504 | Kumamoto University, Japan

RESEARCH INTERESTS

Gravitational wave astronomy using pulsar timing arrays; noise modelling and Bayesian data analysis for PTA datasets; neutron star structure and properties in modified gravity theories; precision pulsar timing with low-frequency radio observations.

EDUCATION

Ph.D. <i>Birla Institute of Technology & Science (BITS), Pilani, Hyderabad Campus, India</i> <i>Thesis: "Advanced Studies on Neutron Stars and Pulsars"</i>	Jan 2018 -- Nov 2023
M.Sc. in Astronomy & Astrophysics <i>Osmania University, Hyderabad, India</i> <i>Thesis: "Parametric study of short-period eclipsing Post Common Envelope Binaries"</i>	Aug 2016 -- Jun 2018
M.Sc. in Mathematics <i>Osmania University, Hyderabad, India</i>	Jun 1998 -- Apr 2000
B.Sc. in Mathematics, Physics & Chemistry <i>Govt. Giriraj College (Osmania University), Nizamabad, India</i>	Jun 1995 -- Apr 1998

RESEARCH POSITIONS

Research Associate <i>Kumamoto University, Kumamoto, Japan</i> - Computational learner modeling for the miXai learn project	May 2026 -- present
Postdoctoral Researcher <i>Kumamoto University, Kumamoto, Japan</i> - Computational learner modeling for the miXai learn project	Dec 2025 -- Apr 2026
Postdoctoral Researcher <i>Kumamoto University, Kumamoto, Japan</i> - InPTA Data Release 2: Dataset processing and timing analysis - InPTA DR2 Noise Analysis: Customised single-pulsar noise modelling - IPTA Data Combination: Contributing to the IPTA DR3	Nov 2023 -- Nov 2025

TEACHING EXPERIENCE

Teaching Assistant: Introduction to Radio Astronomy <i>BITS Pilani, Hyderabad Campus, India</i> - Introduction to Radio Astronomy course for masters students	2022
Teaching Assistant: Introduction to Astrophysics <i>BITS Pilani, Hyderabad Campus, India</i> - Introduction to Astrophysics course for masters students	2022
Teaching Assistant: PHY F110 Physics Laboratory <i>BITS Pilani, Hyderabad Campus, India</i>	2019 -- 2021

- Classical Physics laboratory course for undergraduate students (40 hours per semester)

SELECTED PUBLICATIONS

Full list: ORCID 0000-0003-2715-4504 | 21 peer-reviewed publications

First-Author Papers

- [1] K. Nobleson, C. Dwivedi, S. Desai, et al., "The Indian Pulsar Timing Array data release 2: II. Customised single-pulsar noise analysis and noise budget", J. High Energy Astrophys. 53, 100594 (2026)
- [2] K. Nobleson, S. Banik, T. Malik, "Unveiling a universal relationship between the $f(R)$ parameter and neutron star properties", Phys. Rev. D 107, 124045 (2023)
- [3] K. Nobleson, N. Agarwal, R. Girgaonkar, et al., "Low-frequency wideband timing of InPTA pulsars observed with the uGMRT", MNRAS 512(1), 1234-1243 (2022)
- [4] K. Nobleson, T. Malik, S. Banik, "Tidal deformability of neutron stars with exotic particles within a density dependent relativistic mean field model in R-squared gravity", JCAP 2021(08), 012 (2021)
- [5] K. Nobleson, A. Ali, S. Banik, "Comparison of perturbative and non-perturbative methods in $f(R)$ gravity", Eur. Phys. J. C 82, 32 (2022)

Co-Lead Papers

- [6] P. Rana, P. Tarafdar, K. Nobleson, et al., "The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis", PASA 42, e108 (2025)
- [7] P. Tarafdar, K. Nobleson, P. Rana, et al., "The Indian Pulsar Timing Array: First data release", PASA 39, e053 (2022)
- [8] A.K. Paladi, C. Dwivedi, P. Rana, K. Nobleson, et al., "Multiband extension of the wideband timing technique", MNRAS 527(1), 213-231 (2023)

Selected Co-Author Papers

- [9] J. Antoniadis, et al. (EPTA & InPTA), "The second data release from the European Pulsar Timing Array: III. Search for gravitational wave signals", A&A 678, A50 (2023)
- [10] J. Antoniadis, et al. (EPTA & InPTA), "The second data release from the European Pulsar Timing Array: IV. Implications for massive black holes, dark matter, and the early Universe", A&A 685, A94 (2024)
- [11] G. Agazie, et al. (IPTA), "Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background", ApJ 966(1), 105 (2024)
- [12] A. Srivastava, et al., "Noise analysis of the Indian Pulsar Timing Array data release I", Phys. Rev. D 108, 023008 (2023)
- [13] F. Iraci, et al. (EPTA & InPTA), "Combining the second data release of the European Pulsar Timing Array with low-frequency pulsar data", A&A 704, A109 (2025)
- [14] A. Susobhanan, et al., "pinta: The uGMRT data processing pipeline for the Indian Pulsar Timing Array", PASA 38, e017 (2021)

TALKS & PRESENTATIONS

International Pulsar Timing Array (IPTA) 2025

- "The Indian Pulsar Timing Array Data Release 2: II. Noise Analysis" (Presenter)

32nd Meeting of IAGRG, 2022

- "Tidal Deformability of Neutron Stars in R-Squared Gravity" (Presenter)

IPTA Catchup Meeting, 2022

- "The Indian Pulsar Timing Array: First data release" (Presenter)

Research Facility Training Program, Osmania University, 2022

- "Neutron stars as tools to study Advanced Astrophysics" (Invited Talk)

Science at Low Frequencies VIII, 2021

- "Low-frequency Wideband Timing on uGMRT Data" (Presenter)

27th Intl. Conf. on Advances in Relativity and Cosmology, 2021

- "Tidal deformability of NS in R-squared gravity" (Presenter)

Astronomical Society of India, 2020

- "Neutron Stars with realistic EoS in $f(R)$ theories of gravity" (Presenter)

Compact Stars in QCD Phase Diagram VIII, ICTS Bangalore, 2020

- Attendee

PRIOR EMPLOYMENT

Graphic Designer

May 2005 -- Aug 2016

Deloitte Support Services India Pvt. Ltd., Hyderabad, India

- Designed marketing materials (print and digital) for branding and visual identity
- Mentored junior colleagues and managed designer-client communications

Medical Transcriptionist / Proofreader

Aug 2000 -- May 2005

Vasant Scribes Pvt. Ltd., Hyderabad, India

- Transcribed and proofread medical reports with high accuracy across multiple specialties

PROFESSIONAL MEMBERSHIPS

- Astronomical Society of India (ASI), 2018 -- present
- Indian Pulsar Timing Array (InPTA), 2019 -- present
- International Pulsar Timing Array (IPTA), 2021 -- present